



N O N - N O R M A T I V E

$T = \text{sigma_v} \quad \text{det} = -1 \quad (x, y) \rightarrow (-x, y)$

"This reflects the standard."

Informative / Advisory (NOTE)

Owl Semaphore — Non-Normative

OWL 2 / Reflection State (sigma_v) / Standard Specification

Carey James Balboa

Independent DNS Security Researcher

ORCID 0009-0000-5237-9065 CONCEPT-DOI 10.5281/zenodo.19473697
PUBLISHED-VERSION-DOI 10.5281/zenodo.19474599 (v1.2.0) RC-VERSION-DOI TBD_BY_ZENODO_ON_RELEASE
SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION v2.0.0-rc · LICENSE CC-BY-4.0

BANNER-TUPLE :: STATE=NON-NORMATIVE :: TRANSFORM=T = sigma_v det = -1 (x, y) -> (-x, y) :: QUOTE="This reflects the standard." :: VERSION=v2.0.0-rc :: CONCEPT-DOI=10.5281/zenodo.19473697 :: PUBLISHED-VERSION-DOI=10.5281/zenodo.19474599 :: RC-VERSION-DOI=TBD_BY_ZENODO_ON_RELEASE

NON-NORMATIVE — LAYER PROOF PALETTE

**NON-NORMATIVE — v2 palette**

	NON-NORMATIVE #2F8C8C
	outline / detail #1A1A1A
	background (transparent) #FFFFFF00

Source: NON-NORMATIVE-human-gold-branch-transparent-540.png

Presentation-layer composed badges: per-state palette meander/ring around the V_4 -tested owl-only master. The mathematical master is the owl-only transparent PNG (shown below); the surrounding circle is editorial.

OWL-ONLY MATHEMATICAL MASTER — V_4 -TESTED SOURCE

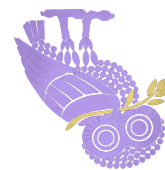
NORMATIVE



NON-NORMATIVE



CRITICAL



METACOGNITIVE

These are the V_4 -tested owl-only PNGs (owl + human-selected gold branch). They are the algebraic master; tests/test_v2_assets.py gates them. The surrounding circle/meander above is presentation only and is not part of the V_4 transform input.

1. Statement of Intent

This document defines the **NON-NORMATIVE owl** as a formal epistemic state within the Owl Semaphore system.

This is not a decorative variation of the normative mark. It is a mathematically defined reflection state with a specific role in analysis, interpretation, and structured disagreement.

The goal is to preserve rigor while allowing controlled deviation from the baseline framework.



2. System Context

The Owl Semaphore system is defined by the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

The NON-NORMATIVE owl corresponds to the vertical reflection operator:

$$\sigma_v : (x, y) \mapsto (-x, y)$$

3. Ontological Role

3.1 Semantic Designation

NON-NORMATIVE represents:

- legitimate alternative interpretation
- reflective critique
- analytical deviation
- structured disagreement

3.2 Interpretive Role

This state indicates that the content:

- departs from the canonical model
- remains structurally grounded in the same system
- is not arbitrary or invalid

3.3 What It Does Not Mean

- not random
- not incorrect by default
- not adversarial (that is CRITICAL)

It is **reflection**, not rejection.

4. Mathematical Definition

4.1 State Operator

$$T_{\text{non}} = \sigma_v$$



4.2 Matrix Form

$$\sigma_v = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

4.3 Determinant

$$\det(\sigma_v) = -1$$

4.4 Properties

- orientation-reversing
- reflection class
- order 2

$$\sigma_v^2 = I$$

5. Coordinate System

The NON-NORMATIVE state uses the same coordinate system as NORMATIVE:

- canvas: 1080×1080
- center: (540, 540)

Transformation is applied relative to this center.

6. Canonical Orientation

6.1 Visual Definition

- upright
- faces LEFT

6.2 Transform Relationship

The NON-NORMATIVE owl is the horizontal mirror of the normative owl.

7. Asset Topology

Layer structure is identical to normative:

- L1 — inner field



- L2 — meander ring
- L3 — owl body
- L4 — outer ring

7.1 Composite Definition

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

8. Geometry

All geometric constraints are inherited from the normative standard:

- identical radii
- identical center
- identical annular structure

No geometric deformation is permitted.

9. Color Specification

9.1 Palette

- outer ring: #316964 (teal)
- owl: #316964 (teal)
- field: #d2d8d6 (cool gray)
- meander: unchanged (gold)

9.2 Color Doctrine

Teal represents analytical distance from the normative baseline while maintaining structural coherence.

10. Transparency and Alpha

Same rules as normative:

- RGBA required
 - corner alpha = 0
 - center alpha = 255
-



11. Provenance

11.1 Construction

Derived from normative by:

- horizontal reflection
- no rotation
- no scaling

11.2 Transform Integrity

The transform must be exact. Any distortion invalidates the state.

12. Asset Invariants

12.1 Algebraic

- operator = σ_v
- determinant = -1

12.2 Visual

- upright
- mirrored orientation

12.3 Structural

- geometry unchanged
 - layer structure unchanged
-

13. Integrity Regime

All assets must:

- pass SHA-3-512 verification
 - be reproducible from layers
-

14. Interpretation Rules



14.1 Positive Rule

When present:

The content reflects the normative framework while offering a structured alternative.

14.2 Negative Rule

It does not indicate failure or error.

15. Non-Permitted Changes

- rotation
 - vertical flip
 - C2 inversion
 - geometry alteration
-

16. Relationship to Other States

- NORMATIVE → identity (*“This is the standard.”*)
 - NON-NORMATIVE → reflection (*“This reflects the standard.”*)
 - CRITICAL → inversion (*“This inverts the standard.”*)
 - METACOGNITIVE → frame-audit (*“The observer audits the frame.”* — thinking examines its own frame)
-

17. Formal Definition

Let L_1 – L_4 be the layer fields.

$$N_{\text{non}} = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

with

$$T = \sigma_v$$

18. Closing Statement

The NON-NORMATIVE owl preserves system integrity while enabling structured deviation.

It is the mechanism by which the system permits disagreement without collapse.



OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER (v2.0.0-rc)



NORMATIVE

$T = I \quad \det = +1$

$(x, y) \rightarrow (x, y)$

“This is the standard.”

RFC 2119 MUST / SHALL



NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \rightarrow (-x, y)$

“This reflects the standard.”

Informative / Advisory (NOTE)



CRITICAL

$T = C_2 \quad \det = +1$

$(x, y) \rightarrow (-x, -y)$

“This inverts the standard.”

RFC 2119 MUST NOT / SHALL NOT



METACOGNITIVE

$T = \sigma_h \quad \det = -1$

$(x, y) \rightarrow (x, -y)$

“The observer audits the frame.”

Epistemic / Framework (META)

Accessibility: color is not the only carrier. State identity is recoverable from color + orientation + textual label (WCAG 2.2 SC 1.4.1).